

230 kV LINE L1 (SUBSTATION A - SUBSTATION B) - PROTECTION SETTINGS CALCULATION SUMMARY (issued 2025-11)

Line data: 80 km, $z_1 = 0.05 + j0.48$ ohm/km $\rightarrow ZL1 = 4.00 + j38.40 = 38.61$ ohm at 84.05 deg (primary).

Instrument transformers (both line ends, per design drawings): CT 1200:5 (CTR = 240), VT 230 kV : 115 V (PTR = 2000).

Secondary conversion factor: CTR / PTR = 0.120.

Zone 1 (forward, instantaneous): 80 % of $ZL1 = 30.89$ ohm primary $\rightarrow 3.71$ ohm secondary.

Zone 2 (forward, POTT keying and permissive trip): 125 % of $ZL1 = 48.26$ ohm primary $\rightarrow 5.79$ ohm secondary.

Zone 3 reverse (Substation B only, blocks echo): must cover the remote zone 2 overreach beyond bus B with a 1.5 margin:

$$1.5 \times (48.26 - 38.61) = 14.48 \text{ ohm primary } \rightarrow 1.74 \text{ ohm secondary.}$$

Maximum torque angle: 84 deg (all phase mho elements). Loop current supervision 50PP = 0.50 A secondary.

Scheme: POTT, phase elements, fiber channel 6.25 ms each direction.

Substation B is the weak-source terminal: echo enabled at B (EDPU 1 cycle = 32 samples, EDUR 2 cycles = 64 samples, EBLK 2 cycles = 64 samples after zone 3 reverse). Echo disabled at A.

Coordination check: for any fault behind Substation B that Substation A's zone 2 can reach, Substation B's zone 3 reverse asserts before the echo pickup expires, so B does not echo for external faults beyond bus B.

Ground protection of L1 is a separate directional earth-fault scheme and is not part of this summary.